



Ultrasonic Radar System

For

Sensor Lab mini Project

**Third Year (Semester-VI) Bachelors in Information Technology,
Autonomous Program**

By

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Certificate

This is to certify that **Prajyot Shinde, Soham Satpute, Laksh Sodhai** have completed the project report on the **Ultrasonic Radar System for Object Detection and Distance Measurement** topic satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of Mr. Abhishek Chaudhari during the year 2024-2025 as prescribed by an Autonomous Institute Affiliated to University of Mumbai

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Abstract

This project focuses on designing and implementing an **Ultrasonic Radar System** using **Arduino Uno**, **HC-SR04 Ultrasonic Sensor**, and **SG90 Servo Motor**. The system scans a **180°** area, detects obstacles, measures distances, and visualizes the data in real-time using **Processing IDE**. The project integrates **embedded systems, sensor interfacing, and graphical visualization**, making it suitable for applications in **security surveillance, obstacle avoidance, and automation**.

Chapter 1

Introduction

1.1 Overview

Radar systems are widely used in navigation, defense, and automation. This project simulates a **low-cost radar system** using ultrasonic waves for object detection. The **HC-SR04 sensor** emits ultrasonic pulses, and the time delay of reflected waves helps calculate distances. A **servo motor** rotates the sensor, enabling **180° scanning**, while **Arduino** processes data and sends it to a **Processing-based GUI** for visualization.

1.2 Objectives

- Design an ultrasonic radar system using Arduino.
- Implement real-time distance measurement with HC-SR04.
- Use a servo motor for angular scanning (0°–180°).
- Develop a graphical radar display using Processing.
- Test the system for accuracy and responsiveness.

Chapter 2

Review of Literature

2.1 Survey

- Existing radar systems use **RF, LiDAR, or ultrasonic sensors**.
- Ultrasonic sensors are **cost-effective** and suitable for short-range detection.
- Prior works highlight the use of **Arduino and Processing** for real-time visualization.

2.2 Research Gap

- Most projects focus on **static sensing**; this project introduces **dynamic scanning**.
- Limited GUI integration in previous works; this project provides **real-time radar visualization**.

Chapter 3

Project Description

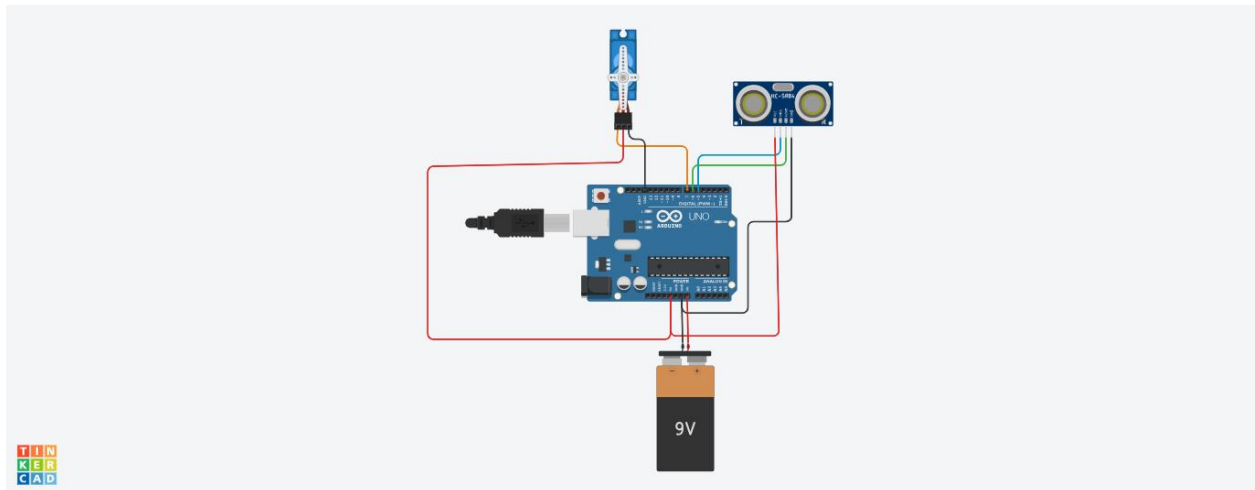
3.1 Problem Definition

- Develop a system to detect objects and measure distances in a **180° range**.
- Provide **real-time visualization** of detected objects.

3.2 Steps Involved

1. **Hardware Setup:** Connect Arduino, HC-SR04, and servo motor.
2. **Software Development:** Write Arduino code for distance calculation and servo control.
3. **GUI Implementation:** Use Processing IDE for radar visualization.
4. **Testing & Calibration:** Validate accuracy and optimize performance.

3.3 Simulation Circuit Diagram



3.4 Component Description

Component	Specifications	Purpose
Arduino Uno	ATmega328P, 16MHz, 5V logic	Main controller
HC-SR04 Sensor	2cm–400cm range, 40kHz pulses	Distance measurement
Servo Motor (SG90)	0°–180° rotation, 4.8V–6V	Sensor rotation
Processing IDE	Java-based GUI development	Radar visualization

3.5 Working Principle

1. **Echolocation:** HC-SR04 emits ultrasonic waves; distance is calculated using:
2. $\text{Distance (cm)} = \Delta t(\mu\text{s}) \times 0.03432$
 $\text{Distance (cm)} = 2\Delta t(\mu\text{s}) \times 0.0343$
3. **Servo Control:** Rotates sensor in **2° increments** for scanning.
4. **Data Visualization:** Arduino sends angle and distance data to Processing for real-time radar display.

Chapter 4

Implementation

4.1 Hardware Setup

- **Connections:**
- Servo Signal → Arduino Digital Pin 7
- HC-SR04 Trig → Digital Pin 5
- HC-SR04 Echo → Digital Pin 6

4.2 Software (Flowchart & Algorithms)

- **Arduino Code:**
- calculateDistance() computes distance using pulse duration.
- Servo sweeps from **15° to 165°** to avoid mechanical stress.
- **Processing Code:**
- Renders radar grid with distance arcs (10cm, 20cm, etc.).
- Displays detected objects as **red dots**.

4.3 Ardino Uno Algorithm:

Start

Include Servo.h library

Define pins:

```
trigPin = 5
echoPin = 6
servoPin = 7
```

Declare variables:

```
duration (for echo time)
distance (for calculated distance)
```

Create Servo object myServo

In $\nabla \rightarrow \leftarrow \blacktriangleleft \blacktriangleright \blacktriangle \blacktriangleright \leftarrow$:

```
Set  $\blacktriangleleft \triangle \downarrow \rightarrow \cup \downarrow \leftrightarrow$  as OUTPUT
Set  $\rightarrow \rightarrow \uparrow \Phi \cup \downarrow \leftrightarrow$  as INPUT
Attach  $\nabla \rightarrow \triangle \blacktriangleleft \Phi \cup \downarrow \leftrightarrow$  to servo
Begin Serial communication
```

In $\blacktriangleleft \Phi \Phi \blacktriangle \blacktriangleright \leftarrow$:

Forward sweep:

For $\rightarrow \rightarrow \rightarrow \rightarrow \blacktriangleleft \rightarrow \uparrow \leftrightarrow \uparrow$ to $\leftrightarrow \downarrow \uparrow$ with step $\uparrow \uparrow$

Rotate servo to $\rightarrow \rightarrow \rightarrow \rightarrow \blacktriangleleft \rightarrow$
Call

$\rightarrow \rightarrow \blacktriangleleft \rightarrow \blacktriangleleft \rightarrow \blacktriangleleft \rightarrow \rightarrow \rightarrow \downarrow \nabla$
 $\blacktriangleleft \rightarrow \rightarrow \rightarrow \rightarrow \blacktriangleleft \rightarrow$

Print angle and distance to
Serial Monitor

Backward sweep:

For $\rightarrow \rightarrow \rightarrow \rightarrow \blacktriangleleft \rightarrow \uparrow \leftrightarrow \uparrow$ to $\leftrightarrow \downarrow \uparrow$ with step $\uparrow \downarrow$

calculateDistance() function:

Set trigPin LOW (2 μ s)

Set trigPin HIGH (10 μ s)

Set trigPin LOW

Read echo time using pulseIn(echoPin,
HIGH)

Calculate distance:

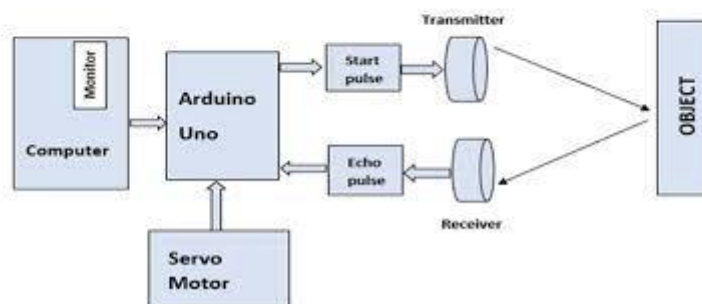
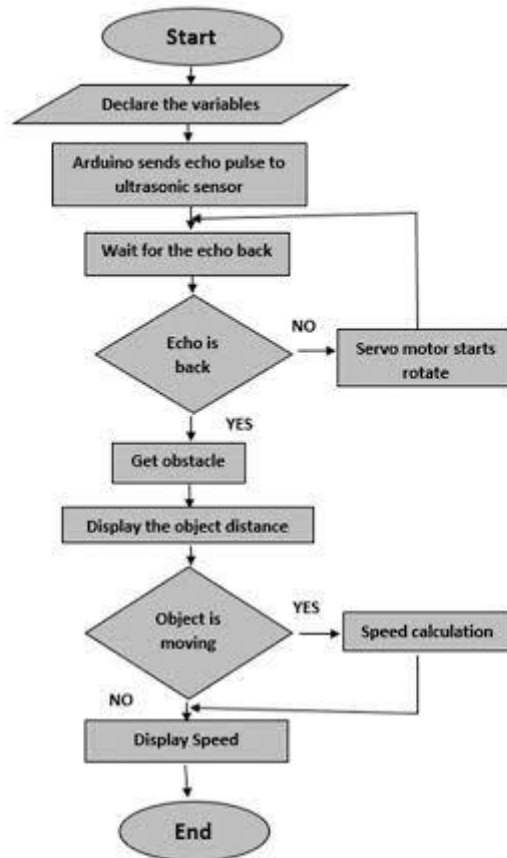
$$\text{Distance} = \frac{\text{Time} \times \text{Speed of Sound}}{2}$$

Return distance

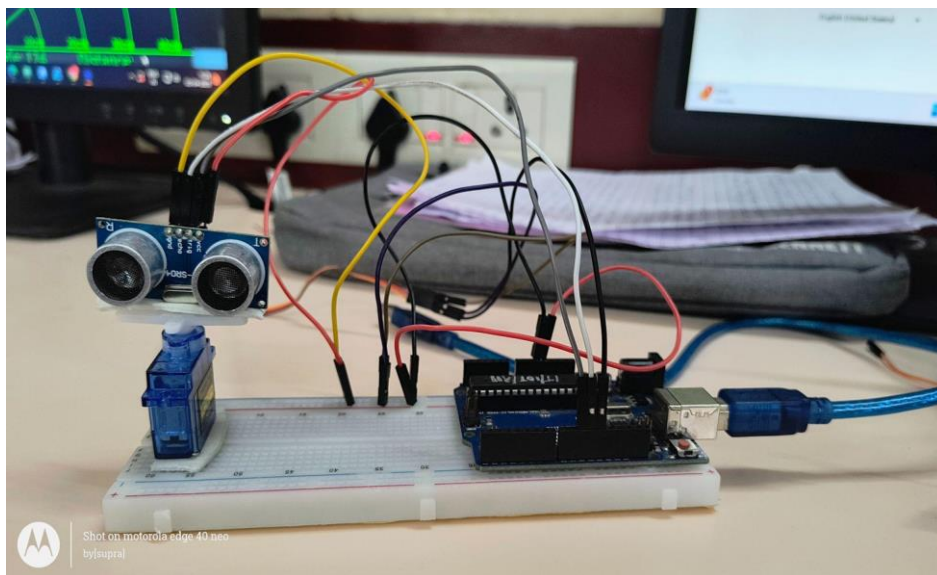
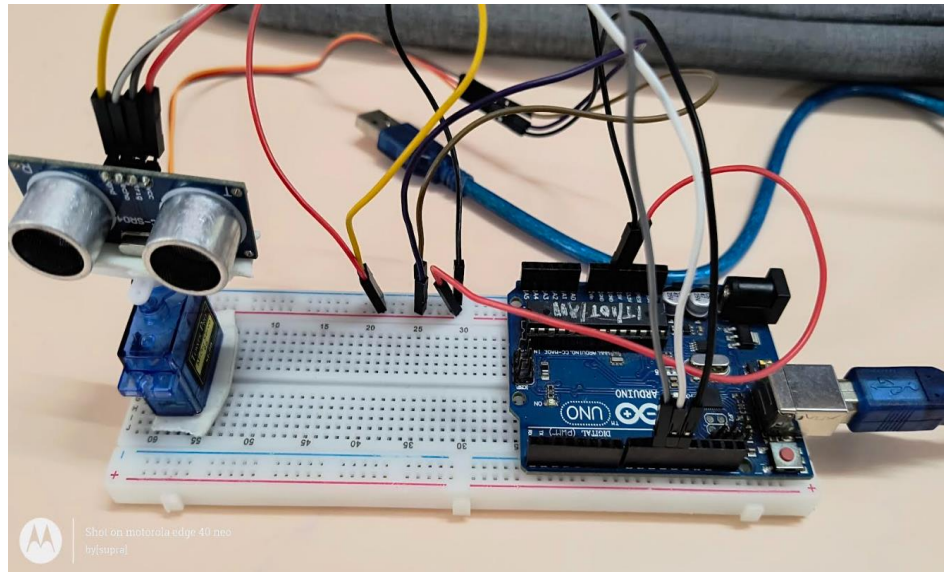
Repeat loop

End

4.4 Flow Chart:



4.5 Project Implementation:

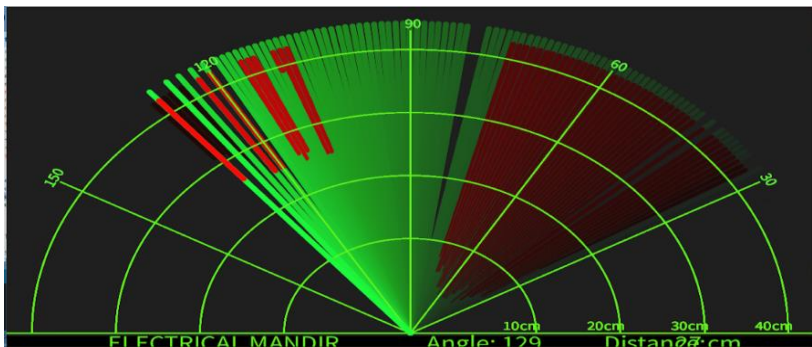


Chapter 5

Results and Conclusion

5.1 Observations

- **Serial Monitor Output:** Displays angle and distance in real-time.
- **Processing GUI:** Shows **green sweep line (current angle)** and **red dots (detected objects)**.



5.2 Performance Metrics

Parameter	Value
Detection Range	2cm–400cm
Angular Resolution	2° steps
Refresh Rate	~5Hz

5.3 Applications

- **Security Surveillance:** Intruder detection.
- **Obstacle Avoidance:** Robotics & autonomous vehicles.
- **Parking Assistance:** Distance measurement.

5.4 Challenges & Solutions

Challenge	Solution
Sensor Noise	Averaging multiple readings.
Servo Jitter	Increased delay (50ms).
Limited Range	Optimized sensor placement.

5.5 Future Enhancements

- **Multiple Sensors:** 360° coverage.
- **Wireless Transmission:** Bluetooth/Wi-Fi integration.
- **Machine Learning:** Object classification.

References

1. Arduino Documentation
2. HC-SR04 Datasheet
3. Processing IDE Tutorials

